MAT 3007 – Optimization

Sensitivity Analysis

Lecture 14

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Announcements:

- The midterm examination will take place on Saturday, March 22nd, 10:00-11:30am in Shaw College F301, F302 and in TC407, TC414.
- You can bring one cheat sheet of A4 paper (with arbitrary notes on both sides of it) in the exam.

  No other (electronic) tools, including calculators, are allowed in the exam.
- **Coverage:** Everything up to sensitivity analysis .

Recap

Production Planning Problem

	Steel	Iron	Copper	Profit
Alloy 1	1	0	1	\$1
Alloy 2	0	2	1	\$2
Resources	100	200	150	

$$
\begin{aligned}
\text{maximize} \quad & x_1 + 2x_2 & & \\
\text{subject to} \quad & x_1 & \leq 100 & \\
& 2x_2 & \leq 200 & \\
& x_1 + x_2 & \leq 150 & \\
& x_1, \quad x_2 & \geq 0. &
\end{aligned}
$$

The objective corresponds to profits; the constraints are resource constraints.

Dual of the Production Planning Problem

Write the dual problem:

$$
\begin{array}{rrrrrl}
\text{minimize} & 100p_1 & +200p_2 & +150p_3 & & \\
\text{subject to} & p_1 & & +p_3 & \geq 1 \\
& & 2p_2 & +p_3 & \geq 2 \\
& p_1, & p_2, & p_3 & \geq 0
\end{array}
$$

What is the meaning of this optimization problem?

- Suppose we want to buy all resources from the company at unit prices p_1, p_2 and p_3. What prices should we offer?

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& p_1, & p_2, & p_3 & \geq 0
\end{array}
$$

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- Suppose we want to buy all resources from the company at unit prices p_1, p_2 and p_3. What prices should we offer?
- One unit of resource 1 and resource 3 can produce one unit of product 1 which is worth \$1. Therefore, if $p_1 + p_3 < 1$, the company rather produces by itself (than agree to sell).

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$$

What is the meaning of this optimization problem?

- Suppose we want to buy all resources from the company at unit prices p_1, p_2 and p_3. What prices should we offer?
- One unit of resource 1 and resource 3 can produce one unit of product 1 which is worth \$1. Therefore, if $p_1 + p_3 < 1$, the company rather produces by itself (than agree to sell).
- The buyer wants to buy the resources for the lowest price and the dual problem finds that price.

Alternative Systems

Consider a set of linear inequalities:

$$\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}.$$

Question: Does this system have a solution?

- If we find a solution, then the solution must exist (a certificate for existence).
- Is there a condition under which the solution must not exist (a certificate for nonexistence)?

Theorem

If we can find a vector $\boldsymbol{x}$ satisfying

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}, \quad \boldsymbol{x} \geq \boldsymbol{0}, \quad \boldsymbol{c}^\top \boldsymbol{x} < 0$$

then there must be no solution to the system $\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$.

Proof. Consider the pair of primal-dual linear programs:

Primal		Dual	
min	$\boldsymbol{c}^\top \boldsymbol{x}$	max	0
s.t.	$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0},\ \ \boldsymbol{x} \geq \boldsymbol{0}$	s.t.	$\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$

Suppose there exists $\boldsymbol{x}$ such that

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}, \quad \boldsymbol{x} \geq \boldsymbol{0}, \quad \boldsymbol{c}^\top \boldsymbol{x} < 0.$$

$\tilde{x} = \alpha x$ with $\alpha > 0$ then $\tilde{x}$ is still feasible. and $c^\top \tilde{x} = \alpha c^\top x \to -\infty$ as $\alpha \to \infty$

Then by scaling this $\boldsymbol{x}$, we can make "$\boldsymbol{c}^\top \boldsymbol{x}$" arbitrarily negative:

$\rightsquigarrow$ The primal problem must be unbounded.

(By weak duality thm, However, for an LP, if the primal problem is unbounded, then the dual problem must be infeasible.) Thus, existence of such an $\boldsymbol{x}$ is a certificate that the linear system $\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$ is infeasible. $\square$

Alternative Systems

One can construct many more pairs of such alternative systems.

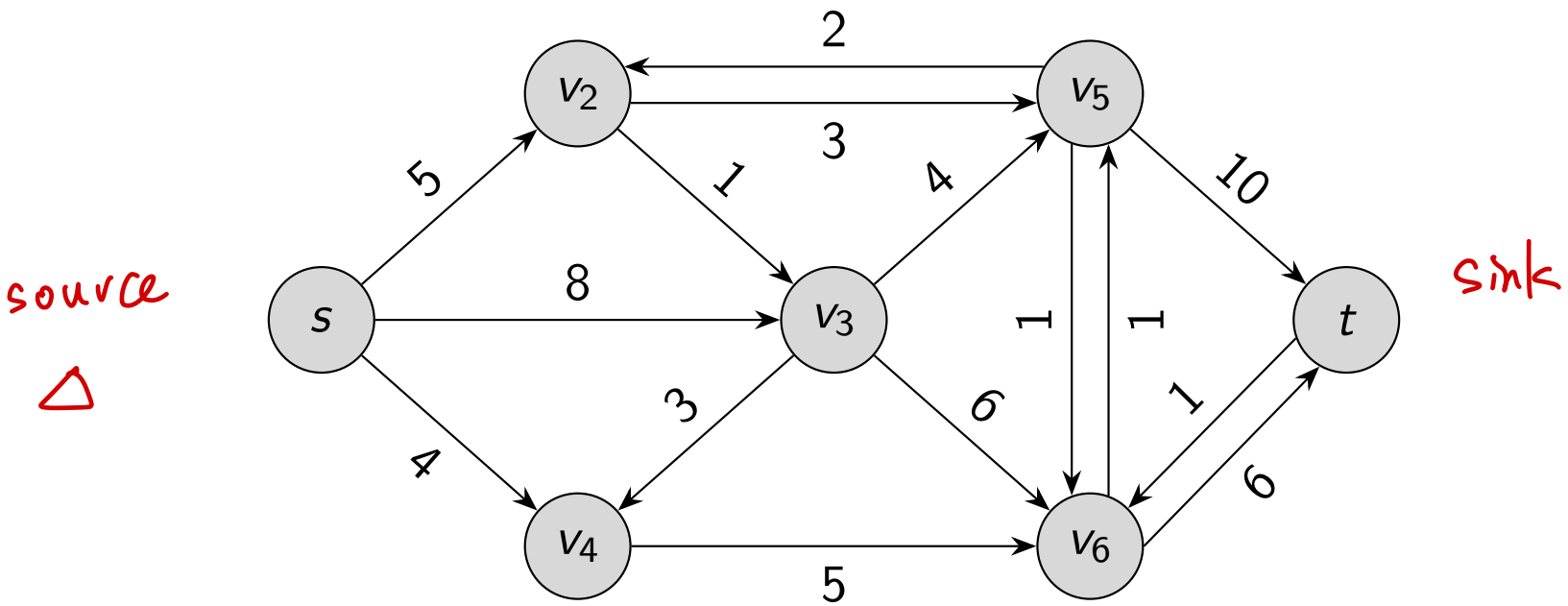
- Typically, it is hard to directly prove that a system is not feasible.
- LP duality provides an alternative approach via considering the dual and checking boundedness.

Maximum Flow Problems

Maximum Flow Problem

The maximum flow problem can be described as follows:

- Given a directed, weighted graph $G = (V, E)$ and a pair of nodes s and t (V: set of nodes; E: set of edges).
- One can think this as a traffic network.
- There is an edge capacity c_{ij} on each edge.
- **Question:** What is the largest amount of flow one can send from s to t subject to the capacity constraints?



LP Formulation

Using this transformation, we can write down the LP formulation. Let x_{ij} denote the amount of flow across edge (i,j), and Δ be amount of flow from s that ends at t.

flow comes from s and ends at t

dual variable

$$\begin{aligned}
\max_{x,\Delta} \quad & \Delta \\
\text{s.t.} \quad & \textstyle\sum_{j:(j,i)\in E} x_{ji} - \sum_{j:(i,j)\in E} x_{ij} = 0, && \forall\, i \neq s,t \quad y_i \\
& \textstyle\sum_{j:(j,s)\in E} x_{js} - \sum_{j:(s,j)\in E} x_{sj} + \Delta = 0 && \quad y_s \\
& \textstyle\sum_{j:(j,t)\in E} x_{jt} - \sum_{j:(t,j)\in E} x_{tj} - \Delta = 0 && \quad y_t \\
& x_{ij} \le c_{ij}, && \forall\, (i,j) \in E \quad z_{ij} \\
& x_{ij} \ge 0, && \forall\, (i,j) \in E.
\end{aligned}$$

flow goes into i = flow goes out of i

flow goes out of s = flow goes into s − Δ

capacity

- The first constraint is the flow balancing constraints for all nodes other than s and t.
- The second (third, resp.) constraint is the flow balancing constraints for node s (t, resp.).

Dual of the Maximum Flow Problem

(Try to derive dual problem?) It's nontrivial.

We construct the dual problem:

min problem pushes z_{ij} to 1 if $y_i = 1, y_j = 0$; $z_{ij} = 0$, o.w.

$$\begin{aligned} \text{minimize} \quad & \sum_{(i,j)\in E} c_{ij} z_{ij} \\ \text{subject to} \quad & z_{ij} \geq y_i - y_j, \quad \forall\, (i,j) \in E \\ & y_s - y_t = 1 \\ & z_{ij} \geq 0 \end{aligned}$$

$\sum c_{ij} z_{ij}$: sum all c_{ij} for $z_{ij} = 1$ $\Rightarrow$ i is in S, j is not in S.

z_{ij}, y_i

If $y_i = 1, y_j = 0$, then $z_{ij} \geq 1$; otherwise, $z_{ij} \geq 0$

$y_s - y_t = 1 \Rightarrow s \in S$, $t \notin S$

say node i is in set S if $y_i = 1$ and not in S if $y_i = 0$

What does the dual problem mean?

First assume all y_i's are either 0 or 1. Then:

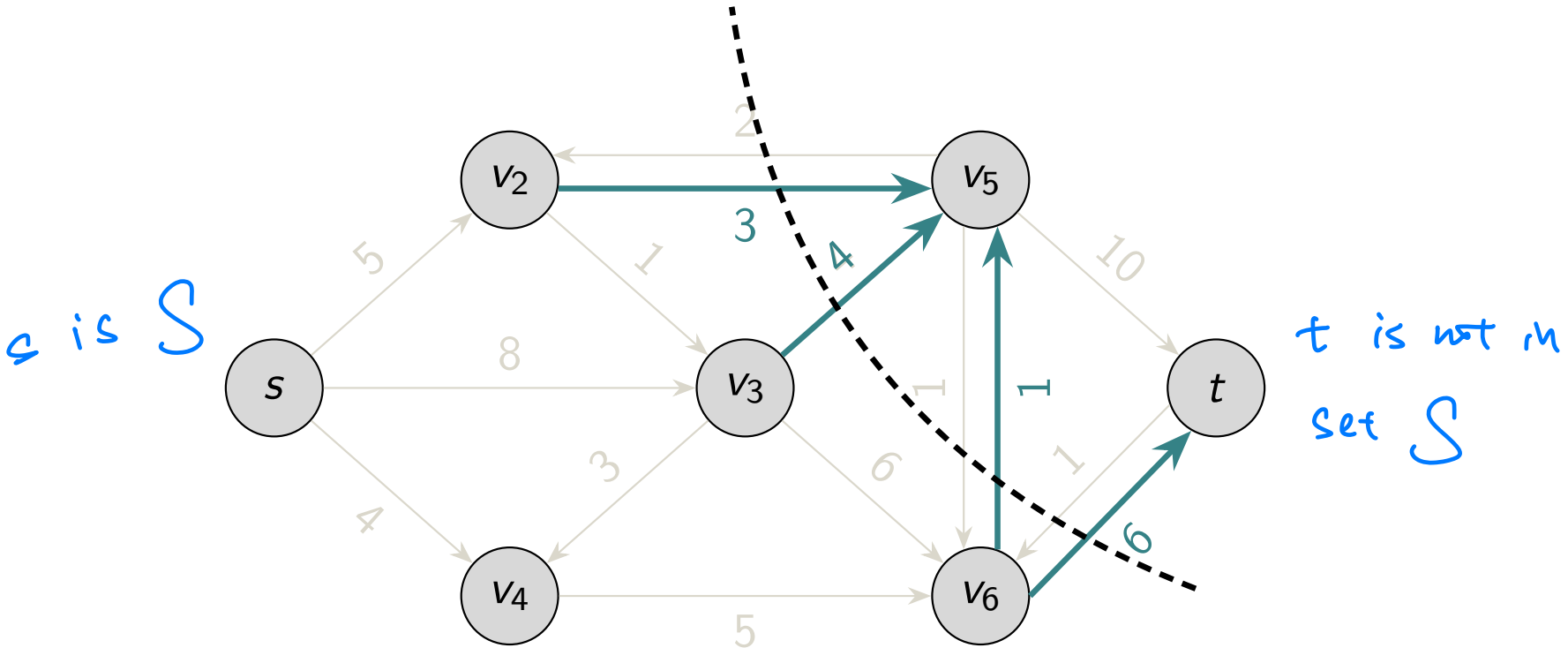
- We assign a label (0 or 1) to each node; 1 to s and 0 to t.
- If i has a larger label than j for $(i,j) \in E$, there is a cost c_{ij}.

Interpretation of the Dual

The dual problem is equivalent to finding a subset S of vertices containing s but not t, that minimizes the weight of a cut, i.e.

$$\sum_{i \in S, j \notin S} c_{ij}$$

- This is called the min-cut problem:



Strong Duality Says ...

Theorem: Max-Flow & Min-Cut

The maximum flow of the network is equal to the smallest cut size of any subset S of vertices.

Corollary

If the value of a flow is equal to the value of some cut, then both are optimal.

- One can view the min-cut as the bottleneck of the network.
- The maximum flow that can be sent through this network is equal to the tightest bottleneck of this network.

This is one classical example of dual problems: maximum flow versus minimum cut.

Sensitivity Analysis

Sensitivity Analysis

One important question when studying LP is as follows:

- How do the optimal solution and the optimal value change when the input changes?

This type of problem is called sensitivity analysis.

- We first study this question from a local perspective and then continue with global discussions.

Consider the standard LP: input (c, b, A)

$$\begin{aligned} \text{minimize}_x \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0}. \end{aligned}$$

We denote the associated optimal value by V.

- If $\boldsymbol{A}$ and $\boldsymbol{c}$ are fixed, V can be viewed as a function of $\boldsymbol{b}$: $V(\boldsymbol{b})$.

Theorem: Differentiability of the Optimal Value Function

If the dual has a unique optimal solution $\boldsymbol{y}^*$, then $\nabla V(\boldsymbol{b}) = \boldsymbol{y}^*$.

- If the dual optimal solution is not unique (or is unbounded or infeasible), then the gradient is not well-defined.
- If one changes b_i by a small amount Δb_i, then the change of the objective value will be $\Delta b_i y_i^*$

Explanation

We know that the optimal value V is also the optimal value of the dual problem:

$$\begin{aligned} \text{maximize}_{\boldsymbol{y}} \quad & \boldsymbol{b}^\top \boldsymbol{y} \\ \text{s.t.} \quad & \boldsymbol{A}^\top \boldsymbol{y} \le \boldsymbol{c}, \end{aligned}$$

change in $V(b)$

$(b+\Delta b)^\top y^* - b^\top y^*$

$= \Delta b^\top y^*$

i.e., $V(\boldsymbol{b}) = \boldsymbol{b}^\top \boldsymbol{y}^*$.

$\rightsquigarrow$ If we change $\boldsymbol{b}$ by a small amount $\Delta \boldsymbol{b}$, such that the optimal sol. does not change, then the change of V must be $\Delta \boldsymbol{b}^\top \boldsymbol{y}^*$.

Similarly, if $\boldsymbol{A}$ and $\boldsymbol{b}$ are fixed, V can be viewed as a function of $\boldsymbol{c}$.

Theorem: Differentiability of $V(\boldsymbol{c})$

If the primal prob. has a unique optimal sol. $\boldsymbol{x}^*$, then $\nabla V(\boldsymbol{c}) = \boldsymbol{x}^*$.

If one changes c_i by a small amount Δc_i, then the change of the objective value will be $\Delta c_i x_i^*$.

- $\rightsquigarrow$ **Reason:** If we change $\boldsymbol{c}$ by a small amount $\Delta \boldsymbol{c}$, such that the optimal solution does not change, then the change of V must be $\Delta \boldsymbol{c}^\top \boldsymbol{x}^*$.

The latter results also hold for inequality constraints (or maximization problems):

$$\begin{aligned} \text{maximize}_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0}. \end{aligned}$$

We have:

1. If the dual has a unique optimal solution $\boldsymbol{y}^*$, then $\nabla V(\boldsymbol{b}) = \boldsymbol{y}^*$.
2. If the primal has a unique optimal solution $\boldsymbol{x}^*$, then $\nabla V(\boldsymbol{c}) = \boldsymbol{x}^*$.

- To see why this must be true, one can add a slack variable and transform it back to the standard form. We can then use the earlier result.

Example — I

$$\begin{array}{lrrl} \text{maximize} & x_1 & +2x_2 & \\ \text{subject to} & x_1 & & \leq 100 \\ & & 2x_2 & \leq 200 \\ & x_1 & +x_2 & \leq 150 \\ & x_1, & x_2 & \geq 0 \end{array}$$

The optimal solution is $\boldsymbol{x}^* = (50, 100)$ with optimal value 250.
The dual optimal solution is $\boldsymbol{y}^* = (0, 0.5, 1)$.

Q1: What is the optimal value if we have 202 units of resource 2? Let's assume the change is "small enough"

- It will change by $\Delta b_2 y_2^* = 1$. Therefore, the optimal value would be 251 ($\rightsquigarrow$ check with CVX: ✓).

Example: Continued — II

$$
\begin{array}{lrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

The optimal solution is $\boldsymbol{x}^* = (50, 100)$ with optimal value 250. The dual optimal solution is $\boldsymbol{y}^* = (0, 0.5, 1)$.

Q2: What is the optimal value if we have 99 units of resource 1?

- It will change by $\Delta b_1 y_1^* = 0$. Therefore, the optimal value would be unchanged ($\rightsquigarrow$ check with CVX: ✓).

Example: Continued — III

$$
\begin{array}{lrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

The optimal solution is $\boldsymbol{x}^* = (50, 100)$ with optimal value 250.
The dual optimal solution is $\boldsymbol{y}^* = (0, 0.5, 1)$.

Q3: What's the opt. value if the profit of product 1 becomes 1.02?

$0.02 \times 50 = 1$

- It will increase by $\Delta c_1 x_1^* = 1$. Therefore, the optimal value would be 251 ($\rightsquigarrow$ check with CVX: ✓).

Example: Continued - IV

$$
\begin{array}{lrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

The optimal solution is $\boldsymbol{x}^* = (50, 100)$ with optimal value 250.
The dual optimal solution is $\boldsymbol{y}^* = (0, 0.5, 1)$.

Q4: What's the opt. value if the profit of product 2 becomes 1.97?

$-0.03 \times 100 = -3$

- It will decrease by $\Delta c_2 x_2^* = -3$. Therefore, the optimal value would be 247 ($\rightsquigarrow$ check with CVX: ✓).

Property: Inactive Constraints

think about production planning

$$\begin{aligned} \text{maximize}_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0} \end{aligned}$$

At an optimal $\boldsymbol{x}^*$ suppose we have $\boldsymbol{a}_i^\top \boldsymbol{x}^* < b_i$. What happens if we change b_i?

$y_i^*(b_i - a_i^\top x^*) = 0$

- By the complementarity conditions, the corresponding dual variable y_i^* must be 0.
- Therefore, changing the right-hand-side of an inactive constraint by a small amount will not affect the optimal value (also the optimal solution).
- **Intuition:** If the stock of a resource is not critical, then increasing/reducing it by a small amount does not matter.

Change of the Optimal Value Function:

- $\nabla V(\boldsymbol{b}) = \boldsymbol{y}^*$, where $\boldsymbol{y}^*$ is the (unique) optimal dual solution.

We call $\boldsymbol{y}^*$ the shadow prices of $\boldsymbol{b}$.

- In the production example, the shadow price of a resource corresponds to the increment of profit if there is one unit more of that resource (locally).

Change of the Optimal Value Function:

- $\nabla V(\boldsymbol{b}) = \boldsymbol{y}^*$, where $\boldsymbol{y}^*$ is the (unique) optimal dual solution.

We call $\boldsymbol{y}^*$ the shadow prices of $\boldsymbol{b}$.

- In the production example, the shadow price of a resource corresponds to the increment of profit if there is one unit more of that resource (locally).
- Therefore, it can be viewed as the unit value for that resource.
- Remember when discussing its dual problem, it stands for unit fair price.

Caveat: How Small is Small Enough?

The above analysis is only local, meaning that it can only deal with small changes.

- It is valid as long as the optimal basis does not change.
- It may not be true otherwise.

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Example: In the production planning problem, if the amount of resource 1 reduces to 0, then the optimal solution will be $(0, 100)$ with optimal value 200 (reduced by 50). This difference would be different from $\Delta b_1 y_1^* = 0$.

$\rightsquigarrow$ We also want to study what ranges of changes belong to small changes.

$\rightsquigarrow$ This is part of the global sensitivity analysis.

Global Sensitivity

Global Sensitivity

We now study what will happen if:

1. $\boldsymbol{b}$ changes to $\boldsymbol{b} + \Delta\boldsymbol{b}$
2. $\boldsymbol{c}$ changes to $\boldsymbol{c} + \Delta\boldsymbol{c}$

We now study what will happen if:

1. $\boldsymbol{b}$ changes to $\boldsymbol{b} + \Delta \boldsymbol{b}$
2. $\boldsymbol{c}$ changes to $\boldsymbol{c} + \Delta \boldsymbol{c}$

Recall the simplex tableau:

$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}$	$-\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{b}$
$\boldsymbol{A}_B^{-1} \boldsymbol{A}$	$\boldsymbol{A}_B^{-1} \boldsymbol{b}$

At the Optimum:

- The reduced costs satisfy $\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A} \geq 0$.
- $\boldsymbol{A}_B^{-1} \boldsymbol{b}$ and $(\boldsymbol{A}_B^{-1})^\top \boldsymbol{c}_B$ are the basic part of the optimal primal solution and the optimal dual solution, respectively.

Changing $\boldsymbol{b}$

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Suppose $\boldsymbol{b}$ becomes $\tilde{\boldsymbol{b}} = \boldsymbol{b} + \Delta\boldsymbol{b}$. Now, the new basic solution corresponding to the original optimal basis is:

$$\tilde{\boldsymbol{x}}_B = \boldsymbol{A}_B^{-1}(\boldsymbol{b} + \Delta\boldsymbol{b}) \overset{B}{=} \boldsymbol{x}_B^* + \boldsymbol{A}_B^{-1}\Delta\boldsymbol{b}.$$

Construct a new solution based on current B.

Note: $\tilde{x} = (\tilde{x}_B, 0)$ is a basic solution by definition

Changing $\boldsymbol{b}$

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$$\tilde{\boldsymbol{x}}_B = \boldsymbol{A}_B^{-1}(\boldsymbol{b} + \Delta\boldsymbol{b}) = \boldsymbol{x}_B^* + \boldsymbol{A}_B^{-1}\Delta\boldsymbol{b}.$$

Note that the reduced costs $\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1}\boldsymbol{A}$ do not depend on $\boldsymbol{b}$.

→ the reduced cost of new $\tilde{x} = (\tilde{x}_B; 0)$ is nonnegative

→ if new $\tilde{x}_B$ is feasible, then it should be optimal

① $A\tilde{x} = \tilde{b}$ → satisfied because $\tilde{x}$ is a BS

② $\tilde{x} \geq 0$ → need to check

Changing b

Suppose $\boldsymbol{b}$ becomes $\tilde{\boldsymbol{b}} = \boldsymbol{b} + \Delta\boldsymbol{b}$. Now, the new basic solution corresponding to the original optimal basis is:

$$\tilde{\boldsymbol{x}}_B = \boldsymbol{A}_B^{-1}(\boldsymbol{b} + \Delta\boldsymbol{b}) = \boldsymbol{x}_B^* + \boldsymbol{A}_B^{-1}\Delta\boldsymbol{b}.$$

Note that the reduced costs $\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1}\boldsymbol{A}$ do not depend on $\boldsymbol{b}$.

- If $\tilde{\boldsymbol{x}}_B \geq 0$, then B is still the optimal basis and the new optimal solution is $(\tilde{\boldsymbol{x}}_B, \boldsymbol{0})$ with the new optimal value:

$$V(\tilde{\boldsymbol{b}}) = \boldsymbol{c}_B^\top \tilde{\boldsymbol{x}}_B = V^* + \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1}\Delta\boldsymbol{b} = V^* + (\boldsymbol{y}^*)^\top \Delta\boldsymbol{b},$$

  where $\boldsymbol{y}^*$ is the optimal dual solution (relate it to the local theorem).

$$c_B^\top \tilde{x}_B = c_B^\top A_B^{-1}(b+\Delta b) = c_B^\top A_B^{-1} b + c_B^\top A_B^{-1}\Delta b$$

original optimal value V^*

- If the original basis is still optimal, then the local sensitivity analysis holds.

Changing b

What ranges of changes qualify for a small change such that the local sensitivity analysis hold? If $\tilde{\boldsymbol{x}} = (\tilde{\boldsymbol{x}}_B; \boldsymbol{0})$ is feasible, then it is still optimal (Why?).

- $\boldsymbol{A}\tilde{\boldsymbol{x}} = \tilde{\boldsymbol{b}}$ still holds since $\tilde{\boldsymbol{x}}$ is a basic solution.
- Need to check $\tilde{\boldsymbol{x}} \geq 0$, i.e., $\tilde{\boldsymbol{x}}_B \geq 0$ $\Rightarrow A_B^{-1}(b+\Delta b) \geq 0$ $\Leftrightarrow A_B^{-1}b + A_B^{-1}\Delta b \geq 0$

Consider the change only occurs in one component of $\boldsymbol{b}$: $\Delta \boldsymbol{b} = \lambda \boldsymbol{e}_i$ ($\boldsymbol{e}_i$ is a vector with 1 at position i). Then, we need to have:

$\Leftrightarrow x_B^* + A_B^{-1}\Delta b \geq 0$

$$\boldsymbol{x}_B^* + \lambda \boldsymbol{A}_B^{-1}\boldsymbol{e}_i \geq 0$$

λe_i

system of inequalities

so that the optimal basis remains the same.

$\rightsquigarrow$ We can find the range of λ by solving these inequalities.

Example: Production Planning

Consider the production example:

$$
\begin{array}{rrrrl}
\text{maximize} & x_1 & +2x_2 & & \\
\text{subject to} & x_1 & & \leq 100 & \\
& & 2x_2 & \leq 200 & \\
& x_1 & +x_2 & \leq \underline{150} & b_3 \\
& x_1, & x_2 & \geq 0 &
\end{array}
$$

The optimal basis is $\{1, 2, 3\}$ and we have $\boldsymbol{x}^* = (50, 100, 50, 0, 0)^\top$.

- How much can we change the third right-hand-side coefficient (150) such that the optimal basis remains the same?

Example: Continued

The final simplex tableau is

B	0	0	0	1/2	1	250
1	1	0	0	-1/2	1	50
3	0	0	1	1/2	-1	50
2	0	1	0	1/2	0	100

Thus $\boldsymbol{A}_B^{-1} = \begin{bmatrix} 0 & -0.5 & 1 \\ 1 & 0.5 & -1 \\ 0 & 0.5 & 0 \end{bmatrix}$.

this part is $A_B^{-1} I = A_B^{-1}$

correspond to slack variables

Example: Continued

The final simplex tableau is

B	0	0	0	1/2	1	250
1	1	0	0	-1/2	1	50
3	0	0	1	1/2	-1	50
2	0	1	0	1/2	0	100

Thus $\boldsymbol{A}_B^{-1} = \begin{bmatrix} 0 & -0.5 & 1 \\ 1 & 0.5 & -1 \\ 0 & 0.5 & 0 \end{bmatrix}$. If $\boldsymbol{b}$ changes to $\boldsymbol{b} + \lambda \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$,

$$\tilde{\boldsymbol{x}}_B = \boldsymbol{x}_B^* + \lambda \boldsymbol{A}_B^{-1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 50 \\ 50 \\ 100 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \geq 0.$$

In order for this to be positive, we need $-50 \leq \lambda \leq 50$.

Changing $\boldsymbol{c}$

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Now suppose $\boldsymbol{c}$ changes to $\tilde{\boldsymbol{c}} = \boldsymbol{c} + \Delta\boldsymbol{c}$. Because feasibility does not depend on $\boldsymbol{c}$, the orginal solution $\boldsymbol{x}^*$ is still a BFS in the new problem.

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- In order for this BFS to be still optimal, we need to guarantee that the reduced costs are nonnegative.
- We only need to consider the non-basic part since the basic part must still be $\mathbf{0}$:

$$\tilde{\boldsymbol{c}}_N^\top - \tilde{\boldsymbol{c}}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N \geq \mathbf{0}.$$

new cost vector

Changing $\boldsymbol{c}$

Now suppose $\boldsymbol{c}$ changes to $\tilde{\boldsymbol{c}} = \boldsymbol{c} + \Delta\boldsymbol{c}$. Because feasibility does not depend on $\boldsymbol{c}$, the orginal solution $\boldsymbol{x}^*$ is still a BFS in the new problem.

- In order for this BFS to be still optimal, we need to guarantee that the reduced costs are nonnegative.
- We only need to consider the non-basic part since the basic part must still be $\mathbf{0}$:

$$\tilde{\boldsymbol{c}}_N^\top - \tilde{\boldsymbol{c}}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N \geq \mathbf{0}.$$

Assume $\Delta\boldsymbol{c} = \lambda \boldsymbol{e}_j$. We discuss two cases: $j \in B$ and $j \in N$. We study how to find ranges for λ such that the original basis is still optimal (so we can apply the local sensitivity analysis).

Case 1: $j \in B$

Assume $j = B(\ell)$. In this case, the reduced costs are:

$$\begin{aligned} & \boldsymbol{c}_N^\top - (\boldsymbol{c}_B^\top + \lambda \boldsymbol{e}_{B(\ell)}^\top) \boldsymbol{A}_B^{-1} \boldsymbol{A}_N \\ = \quad & \boldsymbol{c}_N^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N - \lambda \boldsymbol{e}_{B(\ell)}^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N. \end{aligned}$$

Note that $\boldsymbol{c}_N^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N$ are the reduced costs for the original problem. We denote it by $\bar{\boldsymbol{c}}_N^\top$.

Case 1: $j \in B$

Assume $j = B(\ell)$. In this case, the reduced costs are:

$$\begin{aligned} & \boldsymbol{c}_N^\top - (\boldsymbol{c}_B^\top + \lambda \boldsymbol{e}_{B(\ell)}^\top) \boldsymbol{A}_B^{-1} \boldsymbol{A}_N \\ = \quad & \boldsymbol{c}_N^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N - \lambda \boldsymbol{e}_{B(\ell)}^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N. \end{aligned}$$

Note that $\boldsymbol{c}_N^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N$ are the reduced costs for the original problem. We denote it by $\bar{\boldsymbol{c}}_N^\top$.

Therefore, in order to maintain the optimality of the current basis, we need to have:

$$\bar{\boldsymbol{c}}_N^\top - \lambda \boldsymbol{e}_{B(\ell)}^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N \geq \boldsymbol{0}. \tag{1}$$

Case 2: $j \in N$

Assume $j = N(\ell)$. In this case, the reduced costs are:

$$\boldsymbol{c}_N^\top + \lambda \boldsymbol{e}_{N(\ell)}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N = \bar{\boldsymbol{c}}_N^\top + \lambda \boldsymbol{e}_{N(\ell)}^\top.$$

Therefore, in order to maintain the optimality of the current basis, we need to have:

$$\bar{\boldsymbol{c}}_N + \lambda \boldsymbol{e}_{N(\ell)} \geq \boldsymbol{0}. \tag{2}$$

- We can solve the range of λ from (2).

Example: Production Planning

Consider the same production example:

$$
\begin{array}{rrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

The final simplex tableau is:

B	0	0	0	1/2	1	250
1	1	0	0	-1/2	1	50
3	0	0	1	1/2	-1	50
2	0	1	0	1/2	0	100

How much can we change the first objective coefficient so that we can use the local sensitivity analysis?

Example: Continued

We have $\bar{\boldsymbol{c}}_N = (0.5, 1)^\top$;

read from tableau

$$\boldsymbol{A}_B^{-1}\boldsymbol{A}_N = \begin{bmatrix} -0.5 & 1 \\ 0.5 & -1 \\ 0.5 & 0 \end{bmatrix}.$$

Assume we change the profit 1 from 1 to $1+\lambda$ (i.e., $-1-\lambda$ in the standard form). Then, we need:

since we want to change c_1

$$\bar{\boldsymbol{c}}_N - \lambda \boldsymbol{A}_N^\top (\boldsymbol{A}_B^{-1})^\top \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} - \lambda \begin{bmatrix} 0.5 \\ -1 \end{bmatrix} \geq 0$$

$\rightsquigarrow -1 \leq \lambda \leq 1.$

Example: Continued

We have $\boldsymbol{r}_N = (0.5, 1)^\top$;

$$\boldsymbol{A}_B^{-1} = \begin{bmatrix} 0 & -0.5 & 1 \\ 1 & 0.5 & -1 \\ 0 & 0.5 & 0 \end{bmatrix}; \quad \boldsymbol{A}_B^{-1}\boldsymbol{A}_N = \begin{bmatrix} -0.5 & 1 \\ 0.5 & -1 \\ 0.5 & 0 \end{bmatrix}.$$

Assume we change the profit 1 from 1 to $1+\lambda$ (i.e., $-1-\lambda$ in the standard form). Then, we need:

$$\bar{\boldsymbol{c}}_N - \lambda \boldsymbol{A}_N^\top (\boldsymbol{A}_B^{-1})^\top \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} - \lambda \begin{bmatrix} 0.5 \\ -1 \end{bmatrix} \geq 0$$

$\rightsquigarrow -1 \leq \lambda \leq 1.$

- If the profit coefficient of the first product is between 0 and 2, we can use the local sensitivity theorem to compute the opt. value using $\boldsymbol{x}^*$.

What if the Change is Outside the Range?

If we change $\boldsymbol{c}$ so much such that the reduced cost of the current solution contains negative components, then:

- We can continue with the simplex tableau until it reaches optimal solution.

What if the Change is Outside the Range?

If we change $\boldsymbol{c}$ so much such that the reduced cost of the current solution contains negative components, then:

- We can continue with the simplex tableau until it reaches optimal solution.

If the change of $\boldsymbol{b}$ is so much that the solution corresponding to the original optimal basis B is no longer feasible, then:

- We may need to solve the problem from the start.
- However, we can also have a dual perspective: the objective coefficients of the dual problem have changed. We can then use the method that deals with changes in the objective coefficients.

Other Changes

Changing A

If the change appears in a non-basic column, say in $\boldsymbol{A}_j$, then the original optimal solution is still feasible.

The only change occurs in the reduced costs of jth variable.

- Recompute $\bar{c}_j$. If it is still nonnegative, then the original optimal solution stays optimal. Otherwise, update the tableau for the jth column as well as the reduced cost and continue from there.

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If the change appears in a basic column, then nearly all numbers in the tableau will change. In general, there is not a simple way to deal with it.

Other Changes

Adding a Variable (the rest are kept the same):

- The original BFS is still a BFS, the reduced costs are unchanged.
- We only need to check the reduced cost corresponding to the new variable.
- If it is nonnegative, then the original optimal solution is still optimal; otherwise continue the simplex method from there.

Other Changes

Adding a Variable (the rest are kept the same):

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Adding a Constraint:

- If the original optimal solution satisfies the constraint, then it is still optimal.
- If not, then the best way to deal with it is to interpret it as adding a dual variable. Then use the simplex tableau for the dual problem to continue calculations.

Today's lecture

- Local Sensitivity
- Global Sensitivity

Next lecture

- Interior Point Method

Questions?